MATH 174 - Numerical Analysis

Dani Nguyen

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Contents

1 Error

1.1 Approximation

Let x be an exact number.
Let y be an approximation of x.

Def: The absolute error is $|x - y|$.
Def: The relative error is $\dfrac{|x - y|}{|x|}$,

1.2 Round-off Error

Usually a machine uses a finite amount of memory to store a number.
machine $\leftarrow \pi$
Suppose a machine can store some number of digits.

1. Allocate a digit for sign, positive or negative.
2. Allocate a number of digits for the mantissa.
3. Allocate a digit for sign of the exponent.
4. Allocate a number of digits for the exponent.

Under rounding in the case of a 4 digit mantissa, $\pi = 0.3142 \cdot 10^1$, this is the machine # representation of pi.

Def: Round-off error are all errors arising from the error between actual # and machine #.

Notation:

$$x = \text{real } \#$$
$$fl(x) = \text{machine } \# \text{ representation of } x$$

In an s-digit rounding machine,
Abs error $= |x = fl(x)|$
Rel error $= \dfrac{|x - fl(x)|}{|x|} \leq \frac{1}{2} \cdot 10^{1-5} = 5 \cdot 10^{-5}$

1.3 Floating Point Arithmetic

What happens under arithmetic operations? (+, -, $\cdot$, /)

Two things to watch out for:

1. Many operations can degrade error.
2. Catastrophic cancellation.

1.3.1 Degrading Error

number	machine
x	$fl(x)$
y	$fl(y)$
$x+y$	$fl(fl(x)+fl(y))$

The more nested $fl()$ there are, the larger the error will be. The same follows for -, $\cdot$, /.

When performing arithmetic, it temporarily allows for more digits than the mantissa allows.

Algorithmic ”solutions”

$((x+y)+z)+w$
Worst case, 4 times the error.
$(x+y)+(z+w)$
Worst case, 3 times the error.

1.3.2 Catastrophic Cancellation

Ex: 3-digit rounding machine. Given 2 numbers: 0.312 and 0.311.

$$0.312 - 0.311 = 0.001 = 0.1 \cdot 10^{-2}$$

Loss of significant digits!

Ex: 5-digit rounding machine. Given 2 numbers:

$$\begin{aligned}
x &= 0.41626324 \\
y &= 0.41626323 \\
x - y &= 0.00000001 \\
\text{In machine:} & \\
fl(x) &= 0.41626 \\
fl(y) &= 0.41626 \\
fl(x) - fl(y) &= 0.00000 = 0 \\
fl(fl(x) - fl(y)) &= 0 \\
\text{Relative error:} & \\
\frac{|x - y - fl(fl(x) - fl(y))|}{|x - y|} &= \frac{|x - y - 0|}{|x - y|} \\
&= 1 = 100\% \text{ error!}
\end{aligned}$$

2 Speed

2.1 Dot Product

Consider dot product of vectors

$$\vec{x} \in \mathbb{R}^n, \vec{y} \in \mathbb{R}^n$$

$$\vec{x} \cdot \vec{y} = \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$= x_1 y_1 + x_2 y_2 + \ldots + x_n y_n$$

Count time-consuming operations:

1. Add/subtract
2. Multiply/divide
3. Comparisons
4. Memory access/storage

These are called FLOPS (floating point operations).

For the dot product, there are n multiplications and n-1 additions.
Adding these together, we get $2n - 1$ FLOPS.
Note that this expression is linear in n.

For large n, 2n dominates, $= \mathcal{O}(n) \approx$ constant times n.
ex: For n vs $2n$, $2n$ will take twice as long.
ex: For $\mathcal{O}(n^2)$, n vs $2n$, $2n$ will take four times as long.

2.2 Memory

Consider a matrix $A_{n\times n}$

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} vs \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

The square matrix uses n^2 memory locations, while the vector uses n memory locations. Each number also uses 8 bytes, creating another goal for algorithm development, reduce memory usage.

2.3 MATLAB

A simple MATLAB function to compute the dot product of two vectors and return the number of FLOPS used.

```
function [thedotproduct, theflops] = dotprod(x, y, n)
    thedotproduct = x(1)*y(1);
    theflops = 1;
    for i = 2:n
        thedotproduct = thedotproduct + x(i)*y(i);
        theflops = theflops + 2;
    end
end
```

2.3.1 First Problem

Write a MATLAB function that when given $A\vec{x} = \vec{b}$, computes $\vec{x}$

$$A\vec{x} = \vec{b}$$

A is triangular We assume A is invertible (non-singular) meaning det A = 0